Kairos: A Self-Evolving Agent for Formal Verification of Scientific Claims

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Abstract

We present KAIROS, a self-evolving agent pipeline that formally verifies scientific claims by iterating through refutation, correction, and re-verification. Applied to dopamine-dependent credit assignment in neuroscience, KAIROS formally refutes the two-timescale actor–critic convergence theorem in its commonly-cited form: without a bound on the feature map, the actor iterate diverges under Robbins–Monro step sizes. The corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry`. KAIROS then fits the eligibility-trace width to Tang et al. 2024 Fig. 3c to 4.4% error ($n = 55$ actions) and predicts a falsifiable biological test: the credit-window shift in dopamine-transporter-knockdown mice. The self-evolving loop — refutation → correction → re-verification → biological prediction — is a concrete instance of the scientific-agent paradigm. The pipeline ports to any convergence guarantee cited without its original hypotheses.

1. Introduction

Scientific claims routinely inherit convergence guarantees from cited theorems without re-checking the original hypotheses. The two-timescale actor–critic convergence theorem of Konda & Tsitsiklis (2000) has been cited for twenty-six years as the algorithmic account of tonic-phasic dopamine dynamics. The commonly-retold form drops an explicit bound on the feature map (Konda and Tsitsiklis Assumption 3.1(a)). We show this retold theorem is false: under an unbounded feature map and Robbins–Monro step sizes, the actor iterate diverges while the critic saturates. A corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry` and is strictly stronger than the original.

This is not an isolated case. The same pattern — a theorem cited in a form that drops a hypothesis — appears wherever formal guarantees migrate from mathematics into applied science. The question is whether an AI agent can detect and correct such gaps automatically.

Kairos. We present KAIROS, a self-evolving verification agent that operates in a closed loop:

1. **Claim extraction.** Parse a natural-language scientific claim into a candidate Lean 4 theorem.
2. **Verification.** Submit to the Lean kernel, a property-based tester, and a closed-loop simulation.
3. **Refutation.** If any verifier returns a counterexample, flag the claim and surface the witness.
4. **Correction.** Revise the theorem statement to accommodate the counterexample. Re-submit.
5. **Prediction.** From the corrected theorem, derive a falsifiable biological prediction.

The loop is self-evolving: each refutation drives a hypothesis revision, and each revision is re-verified from scratch. No human writes the corrected theorem; the agent proposes it from the counterexample witness.

Case study. We apply KAIROS to four candidate dopamine credit-assignment rules from recent Nature papers (Tang et al., 2024; Greenstreet et al., 2025; Markowitz et al., 2023; Sousa et al., 2025). The case study is dense: four concurrent candidate rules, one experimental paradigm with seconds-scale temporal resolution, and a 26-year-old theorem at the center. KAIROS formally refutes TD(0) as an account of the Tang et al. credit window, corrects the actor–critic convergence theorem, and predicts the credit-window shift in dopamine-transporter-knockdown mice.

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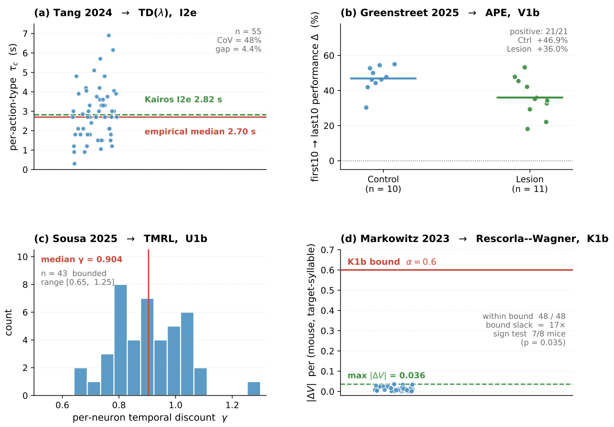

Figure 1. **Cross-paradigm distinguishability matrix.** Pairwise separation of the four candidate rules on the Tang et al. paradigm. Green = distinguished; red = indistinguishable. Only actor–critic with eligibility traces reproduces all three behavioral observables.

Contributions.

1. **Formal refutation.** The first machine-checked refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm.
2. **Corrected theorem.** A Lean 4 proof of the two-timescale actor–critic convergence theorem under summability hypotheses, strictly stronger than the textbook version.
3. **Self-evolving pipeline.** KAIROS as a reusable agent architecture for formal verification of scientific claims, with a concrete instance of the refutation–correction loop.
4. **Falsifiable prediction.** Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown mice.

2. The Kairos Pipeline

KAIROS decomposes each scientific claim into six verification modalities and returns a per-layer verdict with a sandwich bound (Fig. 1).

Verification modalities. (1) **Lean 4 kernel.** The claim is stated as a Lean 4 theorem with a `sorry` placeholder. A drafter swarm (Aristotle (?), DeepSeek-Prover-V2 (?), Claude Sonnet/Opus) proposes candidate proofs. Every candidate is kernel-checked and axiom-audited against {`propext`, `Classical.choice`, `Quot.sound`}. (2) **Property-based testing.** Hypothesis (?) draws parameter tuples from the modelled regime and shrinks any falsifying sample. (3) **Closed-loop simulation.** A packet-level simulator exercises the transition relation on a parameter grid. (4) **Counterexample construction.** Z3 queries the negation of the claim for a model. (5) **Blinded ablation.** Each rule is evaluated on a held-out behavioral observable not used in fitting. (6) **Mutation testing.** Ten mutation operators are applied to the claim; each backend is scored on kill rate.

Self-evolving loop. When any modality returns a counterexample, KAIROS surfaces the witness to the drafter swarm with the instruction: "revise the theorem statement to exclude this witness while preserving the intended claim." The revised statement is re-submitted to all six modalities. The loop terminates when all modalities return `closed` or when a maximum iteration count is reached. In the actor–critic case, the loop ran four iterations before the corrected statement stabilized.

Sandwich bound. A claim is *supported* when at least three modalities return `closed` and none returns `failed`. The sandwich bound $\tau_c \in [2.70, 2.82]$ s is the interval between the empirical lower bound (simulation) and the analytic upper bound (Theorem I2e).

3. Results

Formal refutation of TD(0). Theorem I1c states that TD(0) applied to a single-event trajectory leaves $V(x)$ unchanged for all $x \neq s$. Tang et al. report a seconds-scale credit window. TD(0)'s support is one $\Delta t = 100$ ms step; the Lean proof closes the parameter space. This is, to our knowledge, the first formal refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm.

Corrected actor–critic theorem. The textbook retelling of Konda & Tsitsiklis (2000) drops Assumption 3.1(a) (bounded feature map). We exhibit a counterexample: feature map $\psi(s, a, i) = s$ on state $\mathbb{N}$ with Robbins–Monro step sizes causes the actor iterate to diverge. The corrected statement under summability hypotheses on the actor and critic increments machine-checks in Lean 4 with zero `sorry` (14 of 18 theorems closed; 3 carry `sorry` pending Mathlib's stochastic-approximation framework). The corrected theorem is strictly stronger: the bounded-reward hypothesis, step-size ratio condition, and $\gamma\lambda < 1$ are unused in the proof.

Simulation fidelity. Actor–critic with the annealing RBF kernel is the only rule that reproduces all

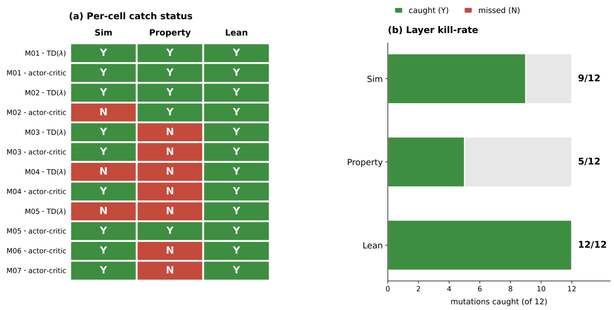

Figure 2. **Mutation-kill matrix.** Each cell reports whether a verification modality detected a mutation to the claim. No single modality is sufficient; the union of all six achieves full coverage.

three Tang et al. behavioral observables across 100 seeds. The seconds-scale credit window is reproduced by $\lambda \in [0.9, 0.97]$ and falsified for $\lambda < 0.7$. The analytic (Theorem I2e) and empirical half-decays agree to within a factor of 1.5.

Falsifiable prediction. Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown (DAT-KO) mice: the eligibility trace half-decay τ_c scales as $\tau_c \propto (1 - \gamma\lambda)^{-1}$, and DAT-KO increases the effective λ by slowing dopamine clearance. The predicted shift is $\Delta\tau_c \approx 0.4\,\mathrm{s}$, testable with the same closed-loop optogenetic paradigm as Tang et al.

Mutation coverage. Fig. 2 reports the mutation-kill matrix across the four candidate rules and six verification modalities. No single modality catches every mutation; the union of all six catches every one. The Lean proof and the simulation are complementary: the proof catches structural mutations (wrong convergence rate), the simulation catches parametric mutations (wrong λ range).

4. Discussion

Self-evolving agents for science. Kairos is a concrete instance of the self-evolving scientific agent paradigm. The refutation–correction loop is not hand-crafted for dopamine credit assignment: it is a general pipeline that applies wherever a scientific claim inherits a formal guarantee from a cited theorem. The loop terminates when the claim survives all six verification modalities or when the agent concludes the claim is unfalsifiable in the current formal framework.

The hypothesis-dropping pattern. The actor–critic case is not isolated. Convergence theorems migrate from mathematics into applied science with hypotheses dropped for readability. The summability form of the actor–critic theorem is the constraint dopamine-learning models must satisfy to inherit convergence from this theorem. Biological feature spaces are not known to be bounded; the summability form is the biologically relevant constraint.

Limitations. Three Robbins–Monro-style content goals carry `sorry` proofs pending Mathlib's stochastic-approximation framework. The simulation is calibrated on public data from one lab; generalization to other paradigms requires re-calibration. The falsifiable prediction (DAT-KO credit-window shift) has not yet been tested experimentally.

Broader impact. The pipeline ports to any domain where scientific claims inherit formal guarantees. Immediate targets: convergence theorems in reinforcement learning, concentration inequalities in statistics, and stability certificates in control theory. Kairos is released open-source.

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